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Department of Computer Engineering
Zakir Husain College of Engineering & Technology
Aligarh Muslim University

Development of A Low-Code Chatbot Builder Tool

Poster No. AI-03

INTRODUCTION

We present a **low-code platform** that empowers users to build intelligent chatbots **without writing a single line of code**. Designed for accessibility, the platform simplifies the process of chatbot development, making it usable by non-programmers and small businesses.
By integrating **large language models (LLMs)** and **retrieval-augmented generation (RAG)**, the platform enables real-time, domain-specific chatbot creation with support for multiple languages and flexible configuration options.

BACKGROUND

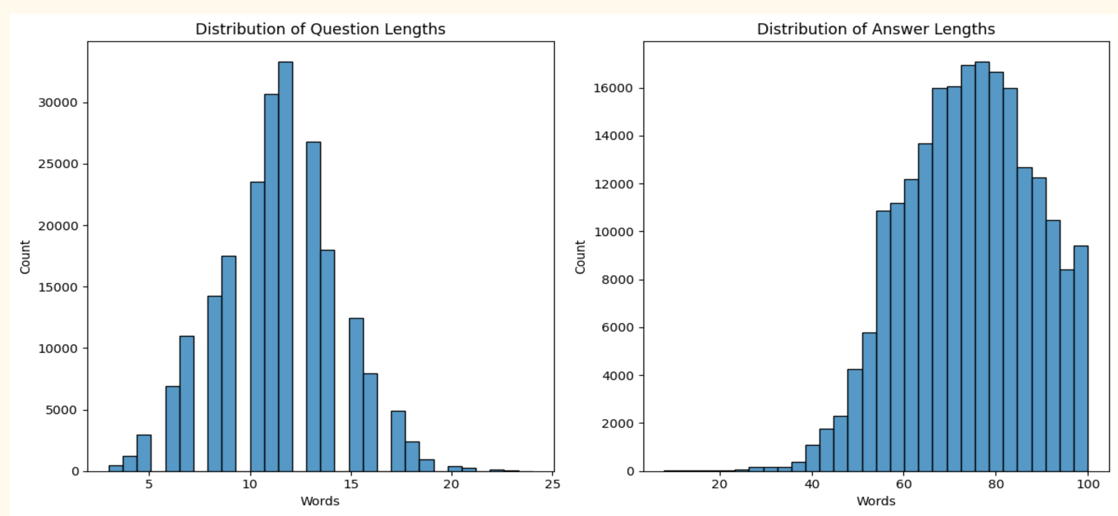
- Most chatbot builders tie users to specific APIs/clouds, limiting flexibility and integration.
- Complex setup and lack of dynamic knowledge, multilingual, or domain customization.
- This project: vendor-neutral, low-code platform for uploading data and custom chatbot settings.
- Enables non-programmers and small businesses to build scalable, accessible conversational AI.

METHODOLOGY

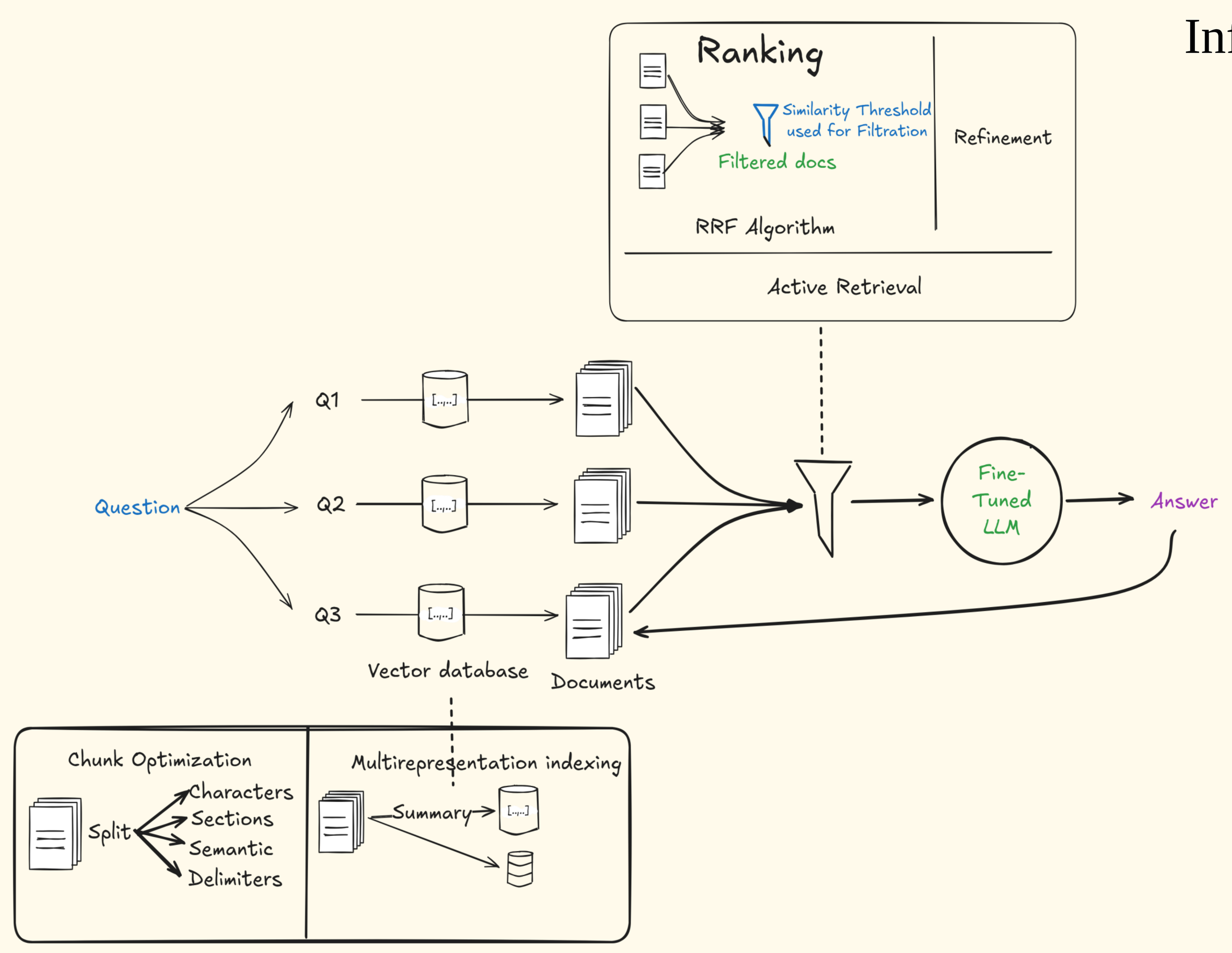
- Fine-tuned LLM on Bitext dataset (~75M tokens) with web scraping, back translation, semantic chunking.
- Preprocessing: data cleaning, LLaMA tokenizer, 80/20 stratified split.
- Trained FLAN-T5 (LoRA) and LLaMA 3.2B (QLoRA) on domain data.
- Compared 4 RAG variants; Fine-Tuned RAG Fusion performed best.
- Used Pinecone for dense retrieval on semantic chunks.
- Integrated RRF and GPT-4o-mini for query expansion.
- Deployed top setup in a low-code chatbot builder with real-time testing, chunk control, and prompt tuning—no coding required.

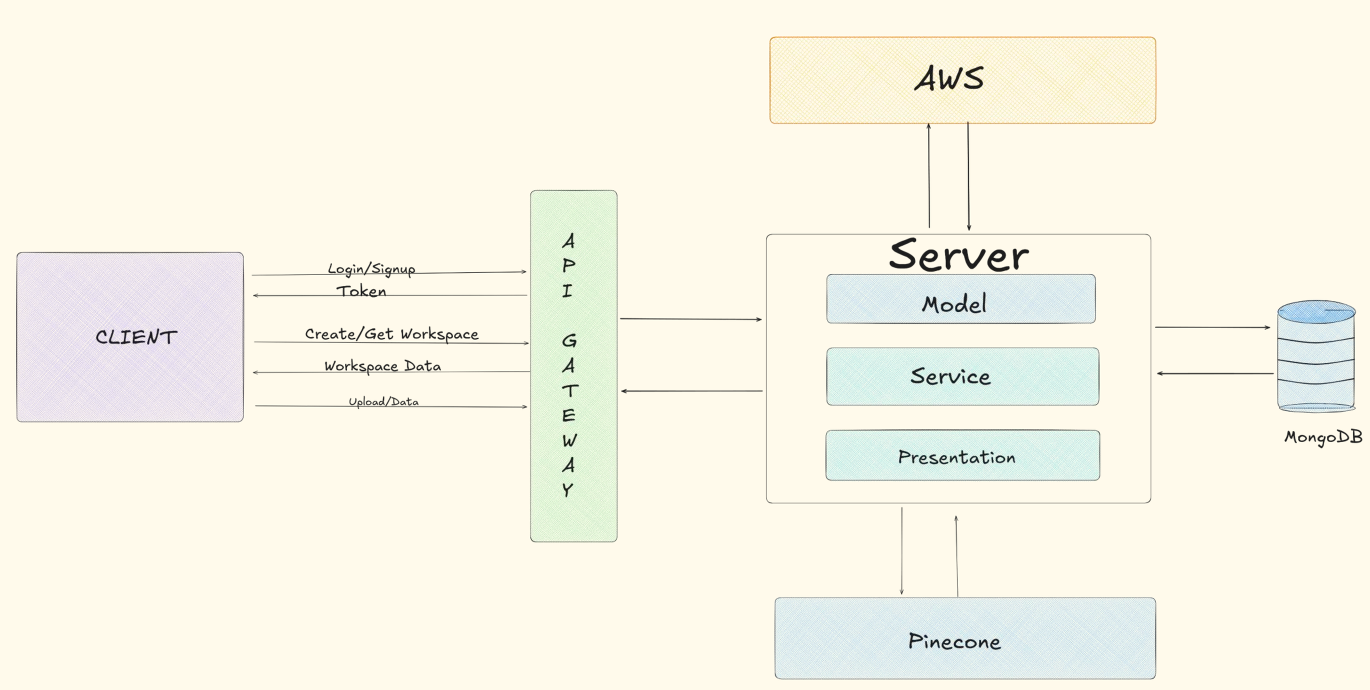
DATASET OVERVIEW

The primary objective was to convert raw multi-domain QA datasets into structured, augmented data suitable for training the RAG Model.
The combined dataset included **345,554 QA pairs** with balanced domain representation and an average context length of approximately **70 words**. It includes data from across 12 domains e.g. Healthcare, IT, Automotive, Finance, E-commerce industries, etc.



ARCHITECTURE & SYSTEM DESIGN





RESULTS

MODEL	ACC(%)	SEM SCORE	G-EVAL	ROUGE-SEM
Google Flan-T5	44	0.28	1.2	0.53
PEFT & QLoRA LLaMA 3 (3.2B)	50	0.59	2.2	0.57
Naive RAG	54	0.70	2.6	0.62
Fine-tuned RAG	66	0.77	2.6	0.66
RAG Fusion	74	0.80	3.2	**0.70**
Fine-Tuned RAG Fusion	**83**	**0.83**	**3.6**	**0.68**

Inference time: **TTFT ≈ 10s**,**TPOT ≈ 2ms**.

CONCLUSION

The **Fine-Tuned RAG Fusion** approach achieved the best results across all evaluation metrics, combining accuracy, scalability, and contextual understanding. Its low-code interface enables users to deploy fast, high-performing chatbots without coding, making AI accessible to non-technical users and small businesses.

REFERENCES

- J. Sánchez Cuadrado, S. Pérez-Soler, E. Guerra, and J. de Lara, "Automating the Development of Task-oriented LLM-based Chatbots," in *Proc. of the 6th ACM Conf. on Conversational User Interfaces (CUI '24)*, Luxembourg, 2024, pp. 1–11. (miso.es)

1. **Imaad Hasan (21AIB120)**
2. **Sayyid Arib Hussain (21AIB386)**